LOCAL FOOD WASTAGE MANAGEMENT SYSTEM

Project Report & Business Documentation

"Connecting Surplus Food Providers to Those in Need"

Domain	**Food Management | Waste Reduction | Social Good**
Tech Stack	**Python | MySQL | Streamlit | Pandas | Plotly**
GitHub	**github.com/itsmanjitdas/local-food-wastage-management-system**
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Date	**June 2026**

Python | SQL | Streamlit | Data Analysis | Social Impact

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1. Executive Summary

The Local Food Wastage Management System is a full-stack data engineering and analytics project that addresses one of the most critical socio-economic challenges of our time — food wastage. Despite a large portion of produced food going to waste, millions of people across the country face food insecurity daily. This system bridges that gap by creating a structured digital platform where surplus food providers (restaurants, supermarkets, grocery stores, and catering services) can list excess food, and receivers (NGOs, charities, shelters, and individuals) can claim those listings before expiry.

> The system was built entirely using Python, MySQL, and Streamlit — demonstrating end-to-end data pipeline capability from raw CSV ingestion, relational database design, SQL analysis, and an interactive web application with CRUD operations and live charts.

Project at a Glance

1,000 Food Providers	**1,000** Receivers	**1,000** Food Listings	**1,000** Claims Tracked
25,794 Total Food Units	**33.9%** Claims Completed	**15+** SQL Queries	**3** App Modules

2. Business Problem & Objectives

2.1 The Problem

Food wastage is a global crisis that operates at a local level. Restaurants prepare excess food, supermarkets discard near-expiry stock, and catering services dispose of large quantities after events. Meanwhile, NGOs, shelters, and individuals in need lack a structured way to discover and claim this surplus food before it perishes. The absence of a centralized platform means valuable food is wasted while hunger persists in the same communities.

2.2 Core Business Objectives

Reduce Food Wastage: Provide a digital registry where surplus food is listed and claimed before its expiry date, reducing unnecessary disposal.

Feed the Needy: Enable NGOs, charities, shelters, and individuals to discover and claim available food efficiently and directly.

Provider Accountability: Track how much food each provider type contributes, enabling data-driven recognition and incentivization.

Geolocation-Based Discovery: Allow receivers to filter food listings by city and location, making discovery fast and relevant.

Data-Driven Decision Making: Generate SQL-powered analytics reports on donation trends, claim rates, and wastage patterns to guide policy and operations.

Operational Transparency: Maintain a real-time claims tracker (Pending / Completed / Cancelled) so all stakeholders can monitor the pipeline end to end.

3. Dataset Overview

The project utilizes four interrelated CSV datasets that were ingested, cleaned, and loaded into a relational MySQL database. Together they form a complete picture of the food donation ecosystem.

Dataset	File	Records	Key Columns	Purpose
Providers	providers_data.csv	1,000	Provider_ID, Name, Type, City, Contact	Entities donating surplus food
Receivers	receivers_data.csv	1,000	Receiver_ID, Name, Type, City, Contact	Entities claiming food donations
Food Listings	food_listings_data.csv	1,000	Food_ID, Food_Name, Quantity, Expiry_Date, Food_Type, Meal_Type	Available surplus food items
Claims	claims_data.csv	1,000	Claim_ID, Food_ID, Receiver_ID, Status, Timestamp	Tracks food claim transactions

3.1 Providers Dataset Profile

Provider Type	Count	Share
Supermarket	262	26.2%
Grocery Store	256	25.6%
Restaurant	246	24.6%
Catering Service	236	23.6%

3.2 Receivers Dataset Profile

Receiver Type	Count	Share
NGO	274	27.4%
Charity	263	26.3%
Shelter	246	24.6%
Individual	217	21.7%

3.3 Food Listings Dataset Profile

Attribute	Details

Total Listings	1,000 food items listed
Total Food Quantity	25,794 units across all listings
Average Quantity/Item	25.79 units per listing
Food Types	Vegetarian (33.6%), Vegan (33.4%), Non-Vegetarian (33.0%)
Meal Types	Breakfast (25.4%), Snacks (25.3%), Lunch (24.8%), Dinner (24.5%)
Top Provider by Volume	Restaurant sector — 6,923 total units donated
Data Period	March 2025

3.4 Claims Dataset Profile

Status	Count	Percentage
Completed	339	33.9%
Cancelled	336	33.6%
Pending	325	32.5%

4. System Architecture & Technology Stack

The project follows a layered data pipeline architecture from raw data ingestion through to interactive visualization, built entirely with open-source Python tools.

Layer	Component	Technology	Role
Data Ingestion	CSV File Loading	Pandas (read_csv)	Load raw datasets into memory
Data Cleaning	Type Conversion & NULLs	Pandas (to_datetime etc.)	Fix dates, handle nulls, standardize
Data Storage	Relational Database	MySQL 8.0	Persistent structured storage with FK constraints
DB Setup	Schema Creation	database_setup.py	Creates all 4 tables with constraints
Data Loading	CSV to MySQL	load_data.py	Inserts all CSV records via mysql-connector
Analysis Layer	15 SQL Queries	MySQL / Pandas read_sql	Aggregations, JOINs, trends, KPIs
Application Layer	Interactive Dashboard	Streamlit + Plotly	3-module web app with filters and CRUD
Version Control	Code Repository	GitHub	Full project hosted publicly

5. Database Design

5.1 Entity-Relationship Summary

The database contains 4 normalized tables with foreign key relationships enforcing referential integrity across the donation-to-claim pipeline.

Table	Primary Key	Foreign Keys	Core Fields
Providers	Provider_ID (INT)	—	Name, Type, Address, City, Contact
Receivers	Receiver_ID (INT)	—	Name, Type, City, Contact
Food_Listings	Food_ID (INT)	Provider_ID → Providers	Food_Name, Quantity, Expiry_Date, Food_Type, Meal_Type, Location
Claims	Claim_ID (INT)	Food_ID → Food_Listings, Receiver_ID → Receivers	Status (Pending/Completed/Cancelled), Timestamp

5.2 Data Flow Pipeline

```
providers_data.csv --> Providers table (1,000 records)
```

```
receivers_data.csv --> Receivers table (1,000 records)
food_listings_data.csv --> Food_Listings table (1,000 records, FK to Providers)
claims_data.csv --> Claims table (1,000 records, FK to Food_Listings + Receivers)
```

> ON DUPLICATE KEY UPDATE is used in load_data.py to ensure idempotent loading — re-running the script will update existing records rather than throw duplicate key errors.

6. SQL Query Analysis — All 15 Business Questions

Each query below directly answers a defined business question. Together they form a complete analytical framework covering providers, receivers, listings, claims, and trends.

Q1: How many food providers and receivers are there in each city?

```
SELECT City, COUNT(DISTINCT Provider_ID) AS Total_Providers, COUNT(DISTINCT Receiver_ID)
AS Total_Receivers FROM ( SELECT City, Provider_ID, NULL AS Receiver_ID FROM Providers
UNION ALL SELECT City, NULL AS Provider_ID, Receiver_ID FROM Receivers ) AS combined
GROUP BY City ORDER BY Total_Providers DESC;
```

Insight: Uses a UNION ALL subquery to merge both tables before grouping by city.

Q2: Which type of food provider contributes the most food?

```
SELECT Provider_Type, SUM(Quantity) AS Total_Food_Donated FROM Food_Listings GROUP BY
Provider_Type ORDER BY Total_Food_Donated DESC;
```

Insight: Restaurants lead at 6,923 units. All four sectors contribute roughly equally (~6,000-7,000 units each).

Q3: What is the contact information of food providers in a specific city?

```
SELECT Name, Type, Contact, Address, City FROM Providers WHERE City = 'Bangalore';
```

Insight: Filter parameter can be swapped to any city. Used by the Browse & Filter module in the app.

Q4: Which receivers have claimed the most food?

```
SELECT r.Name, r.Type, COUNT(c.Claim_ID) AS Total_Claims_Made FROM Receivers r JOIN
Claims c ON r.Receiver_ID = c.Receiver_ID GROUP BY r.Receiver_ID, r.Name, r.Type ORDER BY
Total_Claims_Made DESC LIMIT 10;
```

Insight: JOIN between Receivers and Claims tables, ranked by total number of claim transactions.

Q5: What is the total quantity of food available from all providers?

```
SELECT SUM(Quantity) AS Total_Available_Food FROM Food_Listings;
```

Insight: Result: 25,794 total food units available across 1,000 listings.

Q6: Which city has the highest number of food listings?

```
SELECT Location AS City, COUNT(Food_ID) AS Total_Listings FROM Food_Listings GROUP BY
Location ORDER BY Total_Listings DESC LIMIT 5;
```

Insight: Identifies the highest-demand geographic zones for targeted distribution efforts.

Q7: What are the most commonly available food types?

```
SELECT Food_Type, COUNT(*) AS Listing_Count, SUM(Quantity) AS Total_Quantity FROM
Food_Listings GROUP BY Food_Type ORDER BY Total_Quantity DESC;
```

Insight: Vegetarian, Vegan, and Non-Vegetarian are nearly equally distributed (~33% each).

Q8: How many food claims have been made for each food item?

```
SELECT fl.Food_Name, COUNT(c.Claim_ID) AS Total_Claims_Made FROM Food_Listings fl LEFT
JOIN Claims c ON fl.Food_ID = c.Food_ID GROUP BY fl.Food_Name ORDER BY Total_Claims_Made
DESC;
```

Insight: LEFT JOIN ensures all food items appear even if unclaimed. Reveals high-demand food items.

Q9: Which provider has had the highest number of successful food claims?

```
SELECT p.Name, COUNT(c.Claim_ID) AS Successful_Claims FROM Providers p JOIN Food_Listings
fl ON p.Provider_ID = fl.Provider_ID JOIN Claims c ON fl.Food_ID = c.Food_ID WHERE
c.Status = 'Completed' GROUP BY p.Provider_ID, p.Name ORDER BY Successful_Claims DESC
LIMIT 1;
```

Insight: Three-table JOIN filtered to Completed claims only. Identifies the most impactful provider.

Q10: What percentage of food claims are completed vs. pending vs. cancelled?

```
SELECT Status, COUNT(*) AS Total_Count, ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM
Claims), 2) AS Percentage FROM Claims GROUP BY Status;
```

Insight: Result: Completed 33.9%, Cancelled 33.6%, Pending 32.5% — nearly uniform distribution.

Q11: What is the average quantity of food claimed per receiver?

```
SELECT r.Name, ROUND(AVG(fl.Quantity), 2) AS Avg_Quantity_Claimed FROM Receivers r JOIN
Claims c ON r.Receiver_ID = c.Receiver_ID JOIN Food_Listings fl ON c.Food_ID = fl.Food_ID
GROUP BY r.Receiver_ID, r.Name ORDER BY Avg_Quantity_Claimed DESC;
```

Insight: Measures per-receiver efficiency. High values suggest bulk claimers like NGOs or shelters.

Q12: Which meal type is claimed the most?

```
SELECT fl.Meal_Type, COUNT(c.Claim_ID) AS Claims_Count FROM Food_Listings fl JOIN Claims
c ON fl.Food_ID = c.Food_ID GROUP BY fl.Meal_Type ORDER BY Claims_Count DESC;
```

Insight: Insight drives scheduling — knowing peak meal demand helps providers time their donations.

Q13: What is the total quantity of food donated by each provider?

```
SELECT p.Name, p.Type, SUM(fl.Quantity) AS Total_Donated FROM Providers p JOIN
Food_Listings fl ON p.Provider_ID = fl.Provider_ID GROUP BY p.Provider_ID, p.Name, p.Type
ORDER BY Total_Donated DESC;
```

Insight: Provider leaderboard. Enables recognition programs and targets for underperforming sectors.

Q14: Which food listings expired before being claimed?

```
SELECT fl.Food_Name, fl.Expiry_Date, fl.Location FROM Food_Listings fl LEFT JOIN Claims c
ON fl.Food_ID = c.Food_ID WHERE fl.Expiry_Date < CURDATE() AND (c.Claim_ID IS NULL OR
c.Status = 'Cancelled');
```

Insight: Critical wastage identification query. Flags unclaimed expired items by city for intervention.

Q15: What is the monthly trend of food claims made?

```
SELECT DATE_FORMAT(Timestamp, '%Y-%m') AS Month, COUNT(Claim_ID) AS Total_Claims FROM Claims GROUP BY Month ORDER BY Month ASC;
```

Insight: Time-series trend used to identify peak claim periods. All 1,000 claims fall within March 2025.

7. Streamlit Application — Features & Modules

The app.py file delivers a three-module interactive web application accessible via a sidebar navigation panel. All data is fetched live from the MySQL database on each user interaction.

Module 1: Insights & Trends (Analytics)

- Live KPI metric cards: total providers, receivers, listings, claims, quantity, completed claims
- Exploratory Data Analysis (EDA) expander with descriptive statistics on Quantity column
- Bar chart: Top Contributing Food Providers by sector (Plotly, color-scaled)
- Grouped bar chart: Providers and Receivers breakdown per city
- Pie chart: Top 5 receiver organizations by total claims volume
- Claims status percentage breakdown table (Completed / Pending / Cancelled)
- Interactive SQL Query Vault: dropdown to run any of the 15 queries live with results displayed

Module 2: Browse & Filter Listings

- Loads all Food_Listings records from MySQL on page load
- Three dynamic multi-select filters: City/Location, Food Category (Veg/Non-Veg/Vegan), Meal Type
- Filters apply cumulatively — results update instantly as selections change
- Shows provider contact reference for direct coordination with food donors
- Displays all columns including Expiry Date for urgency-aware claiming

Module 3: Administrative Operations (CRUD)

- CREATE: Form to add new surplus food listing with Food ID, Name, Quantity, Expiry Date, Provider ID
- UPDATE: Form to change claim status (Pending / Completed / Cancelled) by Claim ID
- DELETE: Form to permanently remove a food listing by Food ID with confirmation
- All operations use parameterized SQL queries (prevents SQL injection)
- Success, warning, and error messages displayed after each operation
- Connection is opened and closed safely inside each operation block

> Technical note: All database connections in app.py use mysql.connector with parameterized queries (%s placeholders). Connections are opened within conditional blocks and closed in finally clauses, preventing connection leaks across Streamlit re-renders.

8. Key Insights & Analytical Findings

Provider Sector Balance

All four provider types — Supermarkets (262), Grocery Stores (256), Restaurants (246), and Catering Services (236) — contribute nearly equally to the platform. No single sector dominates, indicating healthy, diverse supply-side participation.

Food Type Distribution

Vegetarian (33.6%), Vegan (33.4%), and Non-Vegetarian (33.0%) food types are almost perfectly balanced across 1,000 listings. This ensures the platform caters to all dietary preferences of receivers.

Meal Timing Insights

Breakfast (25.4%) and Snacks (25.3%) are the most listed meal types, followed closely by Lunch (24.8%) and Dinner (24.5%). This suggests providers across all meal service times are actively participating.

Claims Pipeline Efficiency

Of 1,000 claims tracked, 33.9% are Completed, 33.6% are Cancelled, and 32.5% remain Pending. The high cancellation rate (33.6%) is a critical finding — it signals that nearly 1 in 3 claimed food items is not collected, pointing to a need for better coordination or commitment mechanisms.

Restaurants Lead by Volume

Despite being only the third-largest provider type by count, Restaurants donated the highest total food volume at 6,923 units — suggesting they generate larger per-listing quantities than other sectors.

Receiver Organization Diversity

NGOs (27.4%) and Charities (26.3%) make up the majority of receivers, indicating the platform is primarily serving organized relief operations rather than individual beneficiaries.

Scale of Impact

25,794 total food units are available across the dataset — averaging 25.79 units per listing. If all pending and cancelled claims could be converted to completed ones, the additional food distributed would nearly double the current completed volume.

Expiry Risk

Q14 (expired unclaimed listings) is the most operationally critical query — it identifies wasted food that slipped through the system. This data should drive push notifications and urgency alerts in future system versions.

9. Business Impact & Use Cases

For NGOs and Charities

Search and claim surplus food by city and food type. Coordinate directly with providers via contact information available in the Browse module. Track claim history through the claims table.

For Restaurants and Supermarkets

List surplus food quickly with expiry dates. View real-time analytics on how much their sector is contributing versus competitors. Receive credit for completed claims.

For Municipal Governments and Policy Makers

Use the trend analysis queries (Q6, Q15) to identify cities with highest wastage and lowest claim rates, enabling targeted intervention programs and public awareness campaigns.

For Platform Administrators

Use the CRUD module to maintain data quality — add new provider listings, update stale claim statuses, and remove expired or irrelevant records without needing database access.

For Data Analysts

The 15-query SQL vault provides an analytical framework that can be extended. Expiry analysis (Q14) and monthly trends (Q15) form the basis of predictive models for demand forecasting.

10. Conclusion & Future Scope

The Local Food Wastage Management System successfully demonstrates how a structured data platform can solve a real-world social problem. By combining a relational MySQL backend, a Python-based data pipeline, 15 targeted SQL analytical queries, and an interactive Streamlit web application, the project delivers a functional proof-of-concept for food redistribution at scale.

Future Scope

- Geolocation maps using Folium or Google Maps API to visualize providers and receivers spatially
- SMS/Email alert system to notify receivers when new listings are added in their city
- Machine learning model to predict claim probability based on food type, meal type, and location
- Mobile-first Progressive Web App (PWA) version for on-field access by NGO workers
- Multi-language support (Hindi, Assamese) for broader grassroots adoption
- Expiry countdown alert system — automated notifications for listings within 24 hours of expiry
- Provider rating and reliability score based on claim completion rates
- Cloud deployment with PlanetScale (MySQL) + Streamlit Community Cloud for public access

Project Repository: github.com/itsmanjitdas/local-food-wastage-management-system

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